

CMPUT 267: Machine Learning I · Fall 2026

Estimation and Empirical Risk

Xi Ye

2026-09-17

Recall: Formal Learning Objective

A learner $\mathcal{A}$ maps a random labeled dataset D to a predictor $\mathcal{A}(D)$.

choose $\mathcal{A}$ so that
$$L(f) = \mathbb{E}_{\mathbf{X}, Y}[\ell(f(\mathbf{X}), Y)],$$
$$\mathbb{E}_D[L(\mathcal{A}(D))] \text{ is small.}$$

i For one observed, fixed dataset $\mathcal{D}$, the learner outputs $\hat{f} = \mathcal{A}(\mathcal{D})$. We want $L(\hat{f})$ to be small.

Recall: Empirical Risk Minimization

Because $P_{\mathbf{X},Y}$ is unknown, ERM uses the observed dataset $\mathcal{D}$ to estimate risk before minimizing it over $\mathcal{F}$.

Step 1: Estimation

Estimate $L(f)$ with the empirical risk $\hat{L}_{\mathcal{D}}(f)$.

Step 2: Optimization

Choose $\hat{f} \in \mathcal{F}$ that minimizes the empirical risk.

$$\hat{L}_{\mathcal{D}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i), y_i), \quad \hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_{\mathcal{D}}(f).$$

Today's Focus: Estimation

The first ERM step asks a finite dataset to approximate performance on future examples.

Target

$$L(f) = \mathbb{E}[\ell(f(\mathbf{X}), Y)]$$

Expected loss under the unknown population distribution.

Estimate

$$\hat{L}_{\mathcal{D}}(f)$$

Average loss on the **observed** dataset.



When is empirical risk a reliable estimate of true risk, and how does the dataset size affect it?

Estimation from Sampled Data

Estimation

Use observed data to approximate a fixed but unknown quantity.

Example: We flip for a lot of times. Estimate the probability of heads.

The estimate changes with the sampled data; the target (true value) does not.

An Unfair Coin

Let $X \in \{0, 1\}$ describe one coin flip:

$X = 1$ for heads, $X = 0$ for tails.

Suppose

$$X \sim \text{Bernoulli}(\alpha).$$

And we want to know α (the probability for heads).

Estimate the Mean with a Sample Mean

Flip the coin independently n times:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\alpha).$$

The **sample mean** is

$$\hat{\alpha} = \overline{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Example with $n = 5$:

$$(1, 0, 1, 1, 1) \implies \hat{\alpha} = \frac{4}{5}.$$

An Estimator Is a Random Variable

Before we flip the coin,

$$\hat{\alpha} = g(X_1, \dots, X_n)$$

is a function of random variables, so it is itself random.

After observing the outcome $(1, 0, 1, 1, 1)$, the same expression has the fixed value $4/5$.

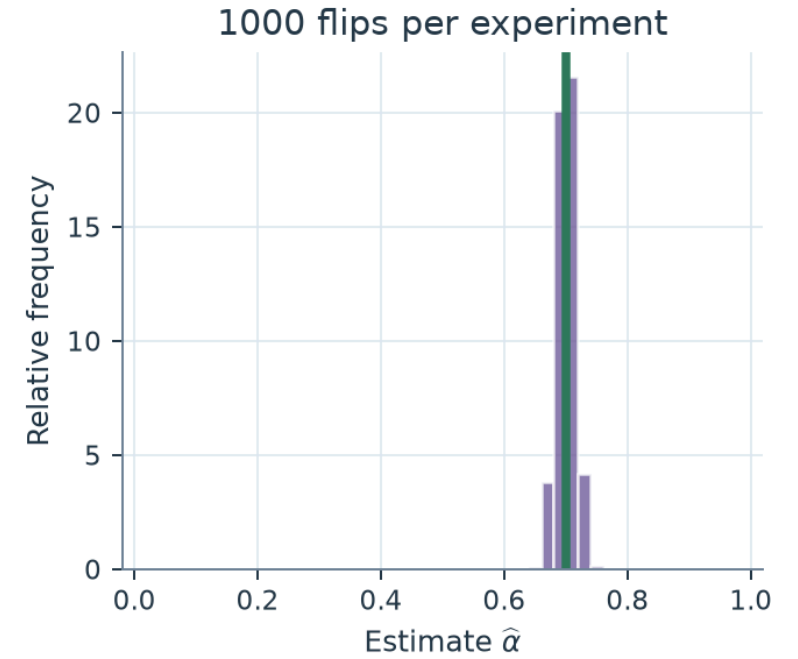
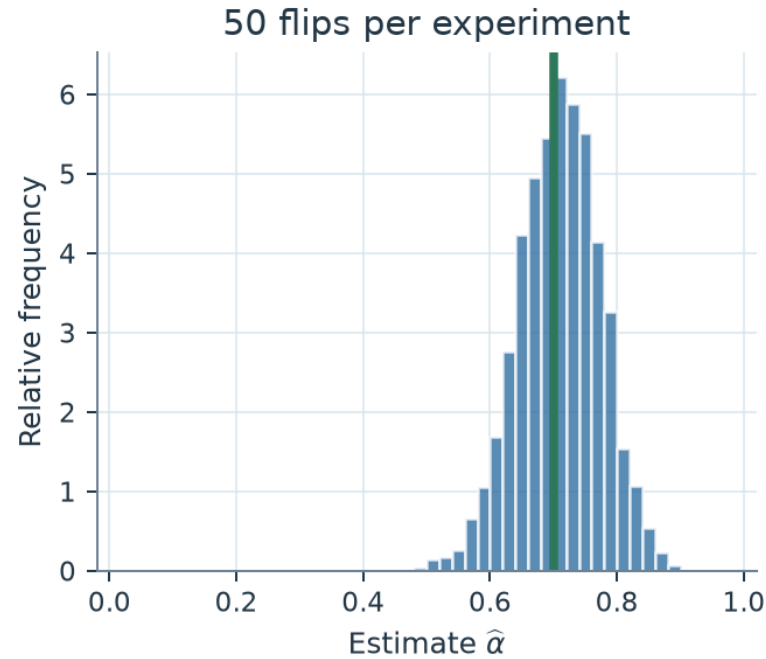
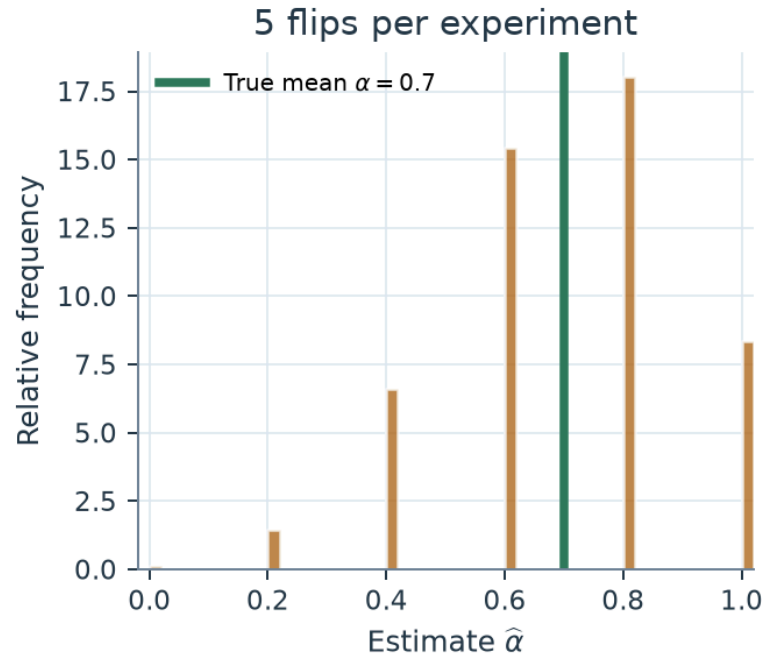
Four Samples, Four Estimates

Suppose the true probability of heads is $\alpha = 4/5$ and each experiment uses $n = 5$ flips.

Experiment	Observed flips	Estimate $\hat{\alpha}$
1	(1, 0, 1, 1, 1)	4/5
2	(0, 1, 0, 1, 1)	3/5
3	(1, 1, 1, 1, 1)	5/5
4	(0, 1, 0, 0, 0)	1/5

Each experiment produces one (random) estimate.

Repeated Samples Give Different Estimates



All three distributions are centered near the same target. More observations make the estimate less variable.

Expectation and Variance of the Estimator

Since $\hat{\alpha}$ is a random variable, what are



$\mathbb{E}[\hat{\alpha}]$ and $\text{Var}(\hat{\alpha})$?

How do they change as the sample size n increases?

Mean of the Estimator

$$\mathbb{E}[\hat{\alpha}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \alpha = \alpha.$$

An estimator is **unbiased** when its expected value equals the quantity it estimates.

i Unbiased does not mean every estimate is correct. It means the estimator is correct **on average over repeated datasets**.

Variance of the Estimator

Recall the variance rules: if A and B are independent,

$$\text{Var}(A + B) = \text{Var}(A) + \text{Var}(B), \quad \text{Var}(cA) = c^2 \text{Var}(A).$$

The flips are independent, so these rules apply to their average:

$$\text{Var}(\hat{\alpha}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n} \text{Var}(X).$$

As the sample size n increases, we have:

- the same **mean** of the estimator: $\mathbb{E}[\hat{\alpha}] = \alpha$
- reduced **variance** of the estimator: $\text{Var}(\hat{\alpha}) = \text{Var}(X)/n$

Sample Mean for I.I.D. Random Variables

Let $Z_1, \dots, Z_n$ be i.i.d. random variables with mean μ and finite variance σ^2 .

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \mathbb{E}[\bar{Z}] = \mu, \quad \text{Var}(\bar{Z}) = \frac{\sigma^2}{n}.$$



We assume the samples in our dataset D are i.i.d.

Risk Is the Mean of a Random Loss

Fix a predictor f . On a random house $(\mathbf{X}, Y)$, define its random loss

$$Z = \ell(f(\mathbf{X}), Y).$$

Its mean is exactly the risk:

$$\mathbb{E}[Z] = \mathbb{E}[\ell(f(\mathbf{X}), Y)] = L(f).$$

Estimating risk is another sample-mean problem: average the observed losses.

Coin Mean and Prediction Risk

Coin Example	Prediction Example
Random flip X_i	Random loss $\ell(f(\mathbf{X}_i), Y_i)$
Unknown mean α	Unknown risk $L(f)$
Estimate $\hat{\alpha} = \frac{1}{n} \sum_i X_i$	Estimate $\hat{L}(f) = \frac{1}{n} \sum_i \ell(f(\mathbf{X}_i), Y_i)$

Empirical Risk Is Average Observed Loss

For an i.i.d. dataset

$$D = ((\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)),$$

the **empirical risk** of a fixed predictor f is

$$\hat{L}_D(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{X}_i), Y_i).$$

A Concrete House Price Estimate

Suppose prices are measured in \$1000s and $f(s) = 80 + 3s$, where $s = lw$ is floor area. For the three houses from Sep 15:

Floor area s_i	Price y_i	Prediction $f(s_i)$	Squared loss
96	310	368	3,364
150	510	530	400
115	355	425	4,900

$$\hat{L}(f) = \frac{3364 + 400 + 4900}{3} = 2888.$$

ERM Chooses Among Candidate Predictors

Consider the small function class

$$\mathcal{F} = \{f_A, f_B\}, \quad f_A(s) = 80 + 3s, \quad f_B(s) = 50 + 3s.$$

Predictor	Empirical risk on the three houses
f_A	2888
f_B	828

❓ Which predictor does ERM choose from $\mathcal{F}$?

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}(f) \quad \implies \quad \hat{f} = f_B.$$

Empirical Risk

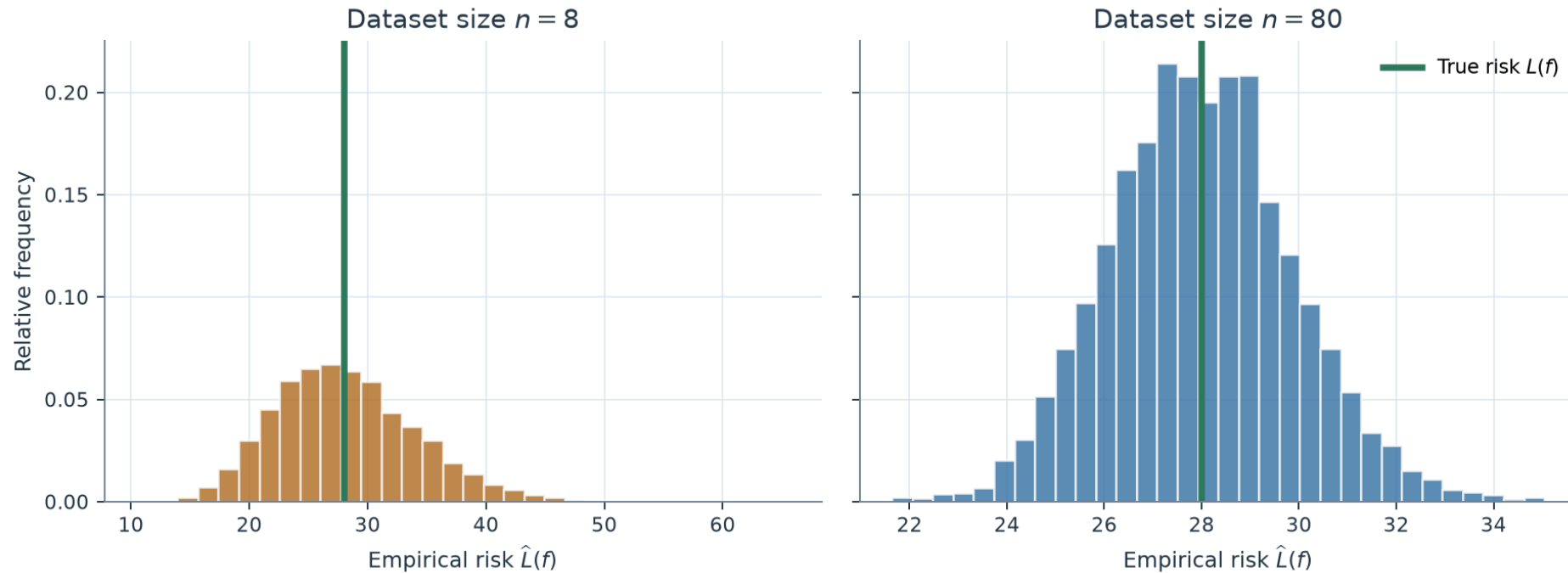
For a fixed predictor f , the observed losses $Z_i = \ell(f(\mathbf{X}_i), Y_i)$ are i.i.d.

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \mathbb{E}[\bar{Z}] = \mu, \quad \text{Var}(\bar{Z}) = \frac{\sigma^2}{n}.$$



Empirical risk is unbiased.
More data reduces the variance.

Larger Datasets Give More Stable Risk Estimates



i The target risk $L(f)$ stays fixed; the empirical estimate changes when the dataset changes.

The ERM Pipeline

Step 1: Estimation

$$L(f) \rightsquigarrow \hat{L}_D(f)$$

Use the dataset to construct an empirical objective.

Step 2: Optimization

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_D(f)$$

Search the function class for a minimizer.

 Today we studied **estimation**. Next we will study how to **optimize** empirical risk.

Caveat

Step 1: Estimation

$$L(f) \rightsquigarrow \hat{L}_D(f)$$

Use the dataset to construct an empirical objective.

Step 2: Optimization

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_D(f)$$

Search the function class for a minimizer.

- ❓ We showed that $\hat{L}_D(f)$ is unbiased when f is fixed before observing D .
ERM uses D to choose $\hat{f}$. Is $\hat{L}_D(\hat{f})$ still an unbiased estimate of $L(\hat{f})$?

Warning

No. The predictor $\hat{f} = \mathcal{A}(D)$ was selected for having small empirical risk on D , so it is no longer independent of the data. As a result, $\hat{L}_D(\hat{f})$ is typically an **optimistic** (biased) estimate of $L(\hat{f})$.

Wrap-up

- Estimation uses data to approximate a fixed unknown quantity.
- The sample mean is unbiased, and its variance decreases like $1/n$.
- For a fixed predictor, empirical risk is an unbiased estimate of true risk.
- More data makes empirical risk less variable.